Student Performance Analysis Using ML and ANN

1. Project Objective

- Analyze student academic performance using data analytics
- Identify factors affecting student results
- Predict Pass or Fail outcome
- Compare Machine Learning and Deep Learning approaches

2. Technologies and Libraries Used

- Pandas for data preprocessing
- Matplotlib and Seaborn for visualization
- Plotly for interactive charts
- Scikit-learn for Machine Learning
- TensorFlow (Keras) for Deep Learning

3. Dataset Description

- Student academic dataset loaded from CSV file
- Includes attendance, study hours, marks, and result status

4. Data Preprocessing

- Missing values handled using mean imputation
- Result column encoded as Pass = 1 and Fail = 0

5. Exploratory Data Analysis

- Correlation analysis performed on numerical features
- Heatmap used to identify important relationships

6. Data Visualization

- Histogram for Final Exam Score distribution
- Box plots to compare attendance and results
- Interactive Plotly visualizations for study patterns

7. Feature Selection

- Attendance Percentage
- Study Hours Per Week
- Assignment Score
- Internal Marks

8. Machine Learning Model

- Logistic Regression algorithm used
- 80-20 train-test data split
- StandardScaler applied for feature scaling
- Model evaluated using accuracy and confusion matrix

9. Deep Learning Model (ANN)

- Artificial Neural Network with one hidden layer
- ReLU and Sigmoid activation functions
- Adam optimizer with Binary Cross-Entropy loss
- Trained for 20 epochs

10. Model Comparison

- Logistic Regression is simple and efficient
- ANN handles complex patterns
- Both models performed effectively

11. Key Observations

- Higher study hours improve performance
- Attendance significantly affects results
- Internal marks influence final outcome

12. Conclusion

- Student performance successfully analyzed
- ML and DL techniques applied effectively
- Helpful for early identification of weak students